



UMBC

Quantum Game Theory

Equilibrium Search, Optimization, and Learning

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Project Framing

We aim to analyze equilibria in classical and quantum strategic games by investigating these theoretical questions:

1. Nash equilibrium after introducing entanglement
2. Eisert-Wilkens-Lewenstein (EWL) protocol
3. Equilibrium search: learning or optimization problem?

Classical Background

Table 1: Possible scenarios from actions {Cooperate, Defect}.

Decision	Outcome
1. Both cooperate	Reward: both banned for 1 month
2. One defects, one cooperates	Temptation: defector receives warning; Sucker's payoff: cooperator permabanned
3. Both defect	Punishment: both banned for 1 year

Classical prisoner's dilemma (cont.)

Ban duration is inversely related to payoff. Define

$$T = 3, \quad R = 2, \quad P = 1, \quad S = 0,$$

so that

$$T > R > P > S.$$

Clyde		Clyde cooperates	Clyde defects
Bonnie			
Bonnie cooperates	$R, 2$	$R, 2$	$S, 0$
Bonnie defects	$T, 3$	$S, 0$	$P, 1$

Figure 1: Payoff matrix for Bonnie and Clyde.

Unlike prisoner's dilemma, a game of chicken contains multiple Nash equilibria and strategic coordination behavior.

$P_1 \backslash P_2$	P_2 swerves	P_2 straight
	P_1 swerves	P_1 straight
swerves	0, 0	-1, 1
straight	1, -1	-1000, -1000

Figure 2: Payoff matrix for the game of chicken played by players P_1 and P_2 .

A strategy profile s is a **Nash equilibrium** if no player can improve by changing strategy alone:

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}).$$

A strategy profile is **Pareto efficient** if no other outcome makes one player better off without making another worse off.

In prisoner's dilemma:

(D, D) is Nash equilibrium,

but

(C, C) is Pareto superior.

This paradox is known as the prisoner's dilemma.

Quantum Computing Background

Quantum computing processes information using principles from quantum mechanics, especially:

- **Superposition**
- **Entanglement**
- **Measurement**

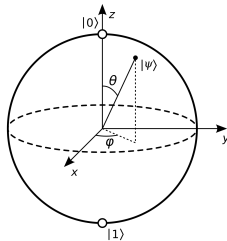
Goal

Quantum game theory studies how quantum phenomena alter strategic decision-making and equilibrium behavior.

0 ●

1 ●

Classical bit stores either 0 or 1



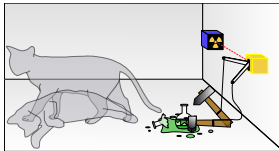
Bloch sphere of a qubit

A qubit may exist in a superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

A quantum state may contain multiple possible outcomes simultaneously until measured, collapsing the state probabilistically:

$|0\rangle$ with probability $|\alpha|^2$, $|1\rangle$ with probability $|\beta|^2$.



Schrödinger's cat: superposition before observation

Interpretation

Quantum systems evolve continuously until measurement.

Two quantum systems are **entangled** when their joint state cannot be separated into independent components:

$$|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle.$$

This means:

- measuring one system reveals information about the other,
- correlations exist beyond classical independence,
- strategies may become correlated through entanglement.

Connection to Quantum Game Theory

Entanglement changes the strategic correlation structure of a game and may alter equilibrium behavior.

Quantum Game Theory

Classically, a player chooses:

C or D .

In the quantum version, the basis states represent classical actions:

$|C\rangle = \text{cooperate}, \quad |D\rangle = \text{defect}.$

A general quantum state may be a superposition:

$$|\psi\rangle = \alpha|C\rangle + \beta|D\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Players do not directly choose outcomes. They apply unitary operations to qubits, and outcomes appear only after measurement.

Eisert, Wilkens, and Lewenstein proposed the following protocol:

1. Start with the classical state:

$$|CC\rangle.$$

2. Apply an entangling operator J .
3. Each player applies a local unitary strategy:

$$U_A \otimes U_B.$$

4. Undo the entanglement with $J^\dagger$, then measure.

The final state is:

$$|\psi_f\rangle = J^\dagger(U_A \otimes U_B)J|CC\rangle.$$

The entangling operator is commonly written as:

$$J(\gamma) = \exp\left(-i\frac{\gamma}{2}D \otimes D\right), \quad \gamma \in \left[0, \frac{\pi}{2}\right].$$

$\gamma = 0 \Rightarrow$ classical game.

$\gamma = \frac{\pi}{2} \Rightarrow$ maximally entangled quantum game.

Interpretation

The parameter γ controls the transition from independent classical choices to correlated quantum strategic behavior.

Classical

(D, D)


 $\gamma = 0$

Transition

correlations emerge



Quantum

(Q, Q)


 $\gamma = \frac{\pi}{2}$

Interpretation

Increasing entanglement changes the strategic correlation structure of the game, shifting equilibrium behavior from classical defection toward cooperative quantum equilibria.

Players choose unitary transformations of the form:

$$U(\theta, \phi) = \begin{pmatrix} e^{i\phi} \cos(\theta/2) & \sin(\theta/2) \\ -\sin(\theta/2) & e^{-i\phi} \cos(\theta/2) \end{pmatrix}.$$

Classical cooperation and defection are special cases:

$$C = I, \quad D = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

EWL also identifies a quantum strategy:

$$Q = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

Players act on qubits in the joint Hilbert space:

$$\{|CC\rangle, |CD\rangle, |DC\rangle, |DD\rangle\}.$$

	$ C\rangle_{P_2}$	$ D\rangle_{P_2}$
$ C\rangle_{P_1}$	$ CC\rangle$	$ CD\rangle$
$ D\rangle_{P_1}$	$ DC\rangle$	$ DD\rangle$

Measurement collapses the final quantum state into one of these classical outcomes.

After measurement, the final quantum state produces probabilities for the four classical outcomes:

$$P_{CC}, \quad P_{CD}, \quad P_{DC}, \quad P_{DD}.$$

Player A's expected payoff is:

$$\mathbb{E}[u_A] = RP_{CC} + SP_{CD} + TP_{DC} + PP_{DD}.$$

Player B's expected payoff is:

$$\mathbb{E}[u_B] = RP_{CC} + TP_{CD} + SP_{DC} + PP_{DD}.$$

For fixed opponent strategy U_B , player A solves:

$$U_A^* \in \arg \max_{U_A \in \mathcal{S}} \mathbb{E}[u_A(U_A, U_B; \gamma)].$$

A Nash equilibrium is a pair (U_A^*, U_B^*) such that:

$$\mathbb{E}[u_A(U_A^*, U_B^*)] \geq \mathbb{E}[u_A(U_A, U_B^*)],$$

$$\mathbb{E}[u_B(U_A^*, U_B^*)] \geq \mathbb{E}[u_B(U_A^*, U_B)].$$

Learning-Theoretic View

The strategy space is continuous and matrix-valued, so equilibrium search resembles nonconvex policy optimization rather than finite action selection.

To make the theory testable, the project can simulate the EWL game over a grid of strategy parameters:

1. Sweep entanglement values $\gamma \in [0, \frac{\pi}{2}]$
2. Grid-search player strategies $U(\theta, \phi)$
3. Compute $|\psi_f\rangle = J^\dagger(U_A \otimes U_B)J|CC\rangle$
4. Convert amplitudes into outcome probabilities:
 $P_{CC}, P_{CD}, P_{DC}, P_{DD}$
5. Plot expected payoffs and approximate best responses

Expected Output

Increasing γ changes best-response behavior and when (Q, Q) becomes stable under the restricted EWL strategy set.

Related Paper Analysis

Main contribution: Meyer showed that a player with access to quantum strategies can outperform a classical opponent in a simple strategic game.

Relevance:

- One of the earliest examples of quantum advantage in game theory.
- Superposition can behave differently from classical randomization.

Relevance

Meyer's paper motivates the idea that strategy spaces can be expanded by quantum mechanics, and introduces the broader question: **what happens when all players have quantum strategies?**

Main contribution: EWL introduced a quantum version of prisoner's dilemma. Under *maximal* entanglement, the classical equilibrium (D, D) can be replaced by the quantum equilibrium (Q, Q) .
At (Q, Q) , both players receive the cooperative payoff

$$(R, R).$$

Interpretation

Entanglement allows stable cooperation *without* communication.

Main contribution: Benjamin and Hayden argued that the EWL result depends strongly on the restricted set of allowed strategies. If players are allowed to use a larger set of unitary operations, then:

- the proposed quantum equilibrium may no longer be stable,
- counter-strategies may exist,
- the cooperative resolution becomes less automatic.

Relevance

This critique turns the project from “quantum games solve prisoner’s dilemma” into a more careful question about assumptions, strategy spaces, and equilibrium stability.

Research Direction

Parameterize each player's strategy by (θ, ϕ) and update parameters using payoff feedback:

$$(\theta_A, \phi_A) \leftarrow (\theta_A, \phi_A) + \eta \nabla_{(\theta_A, \phi_A)} \mathbb{E}[u_A].$$

$$(\theta_B, \phi_B) \leftarrow (\theta_B, \phi_B) + \eta \nabla_{(\theta_B, \phi_B)} \mathbb{E}[u_B].$$

Connection to Statistical Learning

This resembles multi-agent policy-gradient learning: agents iteratively update parameterized strategies using observed payoff signals.

Deliverables

Worst-case: rigorous theoretical comparison and literature review.

Best-case: payoffs and learning trajectories from equilibrium search.

Learning Theory

A player searching for a good quantum strategy is solving an optimization problem:

$$\operatorname{argmax}_{U_A} \mathbb{E}[u_A(U_A, U_B)].$$

In repeated play, agents could update strategies similar to:

- reinforcement learning,
- policy optimization,
- gradient-based learning,
- multi-agent learning.

Interpretation

Quantum equilibrium search can be viewed as learning over a continuous unitary strategy space.

A classical mixed strategy randomizes over actions:

$$pC + (1 - p)D.$$

A quantum strategy transforms amplitudes:

$$|\psi\rangle = \alpha|C\rangle + \beta|D\rangle.$$

The difference is that amplitudes can interfere before measurement.

Theoretical Claim

Unlike classical randomization, entanglement and interference can produce correlations unavailable to independent mixed strategies.

Next computational steps:

- implement payoff-landscape plots over (θ, ϕ, γ) ,
- visualize whether increasing γ changes best-response behavior,
- test when (Q, Q) is stable under the restricted EWL strategy set.

Scope

Although a theoretical project, the next step is connecting the theory via computation and statistical learning.

Limitations and Open Problems

- **Restricted strategy spaces:** EWL's cooperative equilibrium depends on which unitary strategies are allowed.
- **Physical implementation:** Real quantum systems are affected by noise and decoherence.
- **Equilibrium stability:** Enlarging the strategy set may introduce counter-strategies.
- **Learning dynamics:** It is unclear whether adaptive agents would actually converge to the quantum equilibrium.
- **Interpretation:** Quantum advantage may depend on whether entanglement is treated as a resource provided before the game.

Open Question

When does quantum game theory produce new strategic behavior rather than a reformulation of correlated/mixed classical strategies?

Conclusion

The EWL protocol, the first pioneer of quantum game theory, suggests that entanglement can convert the inefficient classical equilibrium of prisoner's dilemma into a cooperative quantum equilibrium.

However, the Benjamin-Hayden critique shows that this conclusion is not unconditional. The stability of the result depends on how the quantum strategy space is defined.

Key Takeaway

Quantum game theory serves as a theoretical bridge between equilibrium analysis, optimization, and learning.

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Thank You!